

19-April-2026

Agenda: Statistics - 10

- p-value
- confidence interval, t-test
- Bayes theorem
- χ^2 distribution & test

p-value

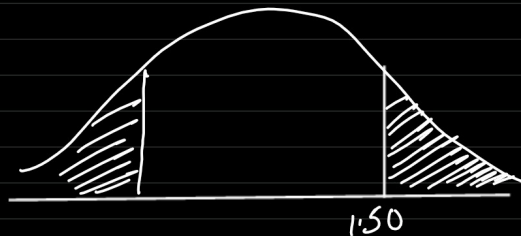
$H_0 \rightarrow \mu = 30 \text{ minutes}$

$H_1 \rightarrow \mu > 30 \text{ minutes} \rightarrow \text{one tail}$

z-stat = 1.50

$\alpha = 0.05$

$P(Z > 1.50)$



$$1 - P(1.50)$$

1 - Area to the left

$$1 - 0.9332$$

$$= 0.0668$$

$$0.0668 < 0.05 \quad \leftrightarrow \quad 6.68\% < 5\%$$

→ we failed to reject H_0

2-tail : —

one-tail : 0.0668

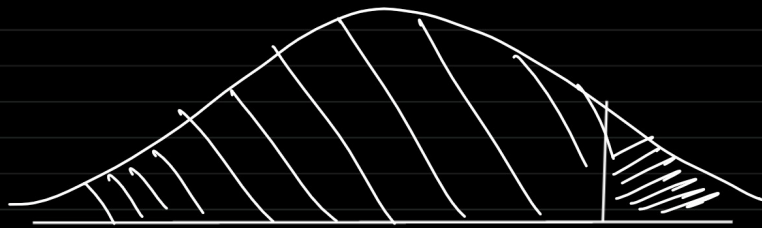
$$\begin{aligned}\text{two-tail p-value} &= \underline{2 \times \text{one-tail p-value}} \\ &= 0.0668 \times 2 \\ &= 0.1336\end{aligned}$$

$$p(z > 1.50) \swarrow$$

$$p(z < -1.50)$$

$$0.1336 < 0.05$$

$$\alpha = 0.05$$



left-danger-zone $\rightarrow \alpha/2$

right danger-zone $\rightarrow \alpha/2$

formula

$$p(z > |z_{\text{observed}}|) < \frac{\alpha}{2}$$

in math:

inequality

$$a < b$$

multiply by 2

$$2a < 2b$$

apply that to our intuition:

$$\text{one-tail probability} < \frac{\alpha}{2}$$

$$\begin{array}{l} \text{multiply both} \\ \text{sides by 2} \end{array} \rightarrow 2 \times \left(\frac{\text{one tail prob}}{p} \right) < 2 \times \frac{\alpha}{2}$$

$$\text{Two tail prob} < \alpha$$

$$2\alpha p < \alpha$$

$$\rightarrow 2p < \alpha$$

$$\rightarrow p < \alpha/2$$

} two tail

$$p < \alpha$$

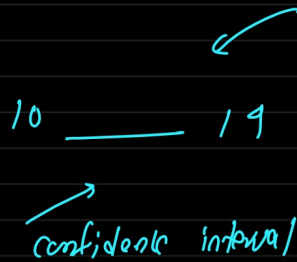
Confidence interval

How long does it take to walk in the park

A $\rightarrow$ exactly 12 minutes $\leftarrow$ point estimate

B $\rightarrow$ Between 10 & 14 minutes $\leftarrow$ range / interval

out of A & B which one is more likely to be "right"



confidence interval (CI) : how wide we make our buffer to be sure to catch the truth inside it.

in stats, $\rightarrow$ we never truly know the true mean (μ)
 $\rightarrow$ we only have sample mean ($\bar{x}$)

CI is a range of values that we are $X\%$ certain contain the population mean

$\rightarrow$ 95% chance that mean is in this specific range, it means if we repeat the experiment 100 times, 95 of the range we calculated would successfully trap the real mean.

Formula:

$$CI = \bar{x} \pm (\text{critical } z \times SE)$$

$\rightarrow x + \text{margin of error} \rightarrow [x, y]$
 $\rightarrow x - \text{margin of error} \rightarrow [x, y]$

numerical example:

$\alpha = 0.05$, 95% confidence interval

$\bar{x} = 31$ $n = 36$, confidence level = 95%.

$\sigma = 4$

step 1: $SE = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{36}} = \frac{4}{6} = 0.6667$

step 2: critical z: 95% confidence (two tail)

critical $z = 1.96$

step 3: ME:

$$ME = 1.96 \times 0.6667 \\ = 1.3067$$

step 4: create the range:

lower band : $31 - 1.3067 = 29.6933$

upper band : $31 + 1.3067 = 32.3067$

95% interval confidence $\rightarrow [29.6933, 32.3067]$ minutes

$H_0: \mu \rightarrow 30$ minutes

if claim value falls inside the interval, you fail to reject the H_0 .

$\mu \rightarrow ?$ $\sigma = ?$ population SD

 $\overline{x} \rightarrow$

The problem: You don't know the true deviation.

The solution:

sample group $\rightarrow \bar{x} \rightarrow$ sample mean
 $\rightarrow (s) \rightarrow$ sample deviation

T-Test :

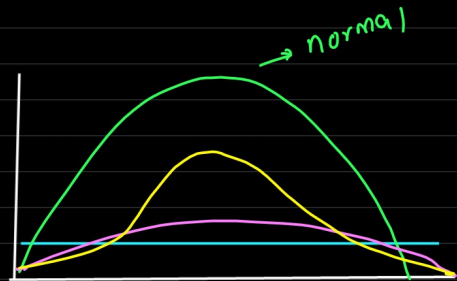
7-test is just a "continuous" z-test, sample deviation instead of population deviation.

(student T-distribution
t-distribution)

1e)) number of data point (n) $\rightarrow$ 1 $\rightarrow$ flat
 $n = 15$ scatter $\rightarrow$ better

$n=30 \rightarrow$ 1er fact

$n = 300 \uparrow \rightarrow$ normal distribution


$$n < 30$$
$$\sigma = ?$$
$$n \uparrow \rightarrow ND$$
$$\eta \rightarrow 1$$
$$n \rightarrow 2$$
$$n=3$$
$$n \rightarrow 4$$
$$n \rightarrow 5$$
$$n : 1000$$

$n: 2000$

✓

ND

T-dist & Degree of freedom,

$$\sigma \rightarrow \infty$$

$$\text{sample SD} \rightarrow (s)$$

$$\downarrow$$
$$\text{estimate} \rightarrow \text{Bell curve} \rightarrow n$$

$$n \uparrow \rightarrow \text{estimate} \uparrow$$

$$df = \underline{\underline{n-1}}$$

$$df \uparrow \rightarrow \text{estimate} \uparrow$$

Formula:

$$\text{z-statistic (z-test)} : \quad z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$\text{T-statistic (T-test)} : \quad t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

problem statement:

scenario: The lightbulb company, they claim their bulb, last 1000 hrs. ($\mu = 1000$)

test:

$$n = 5$$

Data (hours) : 980, 960, 1020, 940, 950

$$\alpha = 0.05$$

$$\underline{\text{step 1:}} \quad \bar{x} = \frac{980 + 960 + 1020 + 940 + 950}{5} = 970 \text{ hours}$$

step 2: calculate sample SD(s)

Distance from mean(970): 10, -10, 50, -30, 20

Square : 100, 100, 2500, 900, 400

Sum : 4000

Divide by $n-1$: $\frac{4000}{5-1} = 1000$

Square root : (s) = $\sqrt{1000} \approx 31.62$

Standard Error (SE) = $\frac{s}{\sqrt{n}} = \frac{31.62}{\sqrt{5}} \approx 14.14$

calculate: t-statistic

$$t = \frac{\bar{x} - \mu}{SE} = \frac{970 - 1000}{14.14} = \underline{\underline{-2.12}}$$

critical T for $df = 4$, $\alpha = 0.05$: $\rightarrow$ T-table

our result : $|-2.12|$

our result < critical t-value, we fail to reject H_0

2.12 < 2.776 $\rightarrow$ fail to reject H_0

$\downarrow$
not enough evidence

Bayes theorem:

we start with an opinion, we update that opinion when we see new evidence.

A → toppe.

→ fail → Qul2 →

→ fails → 5 times →

likelihood
(if your belief is true, how likely is the new evidence)

prior
(initial belief before seeing the new data)

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

posterior (updated belief after seeing the data)

evidence

Scenario:

There is a rare disease then only 1% of population has

$$P(\text{Disease}) = 0.01$$

Medical test:

* If you have the disease, the test is 99% accurate.

positive → 99%.

* If you don't have the disease, the test is 95% accurate

negative → 95%.

positive $\rightarrow 99\%$

negative $\rightarrow 95\% \rightarrow$ false positive rate
 $\downarrow$
 5%

100 $\rightarrow 95\%$ true $\checkmark$
 $\rightarrow 5\%$ false $\rightarrow 5$

step 1: Identifying the "prior" $\rightarrow$ what we know before the test

$$P(A) \text{ [Has Disease]} = 0.01 \text{ (1\%)}$$

$$P(\text{not } A) \text{ [Healthy]} = 0.99 \text{ (99\%)}$$

step 2: Identifying the likelihoods (How the test behaves)

$$P(B|A) \text{ [Test positive if sick]}$$

$\swarrow$ $\uparrow$
positive sick

$$P(B|A) = 0.99 \text{ (99\%)}$$

$$P(B|\text{not } A) = 0.05 \text{ (5\%)}$$

step 3: Total Evidence $P(B)$

$\rightarrow$ total prob of anyone getting a positive test

$\rightarrow$ sick people getting positive $\rightarrow 0.01 \times 0.99 = 0.0099$

$\rightarrow$ Healthy people getting false positive $\rightarrow 0.99 \times 0.05 = 0.0495$

$$\begin{aligned} P(B) &= 0.0099 + 0.0495 \\ &= 0.0594 \end{aligned}$$

$$P(\text{sick} | \text{positive})$$

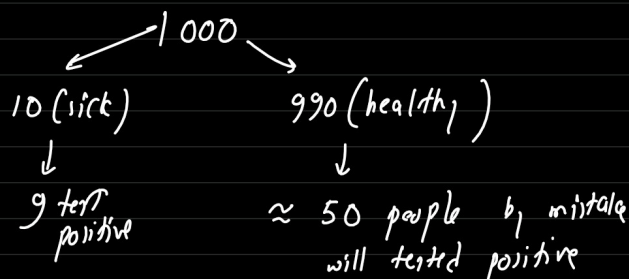
$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

$$P(\text{sick} | \text{positive}) = \frac{P(\text{positive} | \text{sick}) \times P(\text{sick})}{\text{Total evidence}}$$

$$= \frac{0.99 \times 0.01}{0.0599}$$

$$= 0.1667 \rightarrow 16\%$$

Because the disease is so rare (1%) & the false positive rate is 5%, there are actually more healthy people getting false positive.



doctors call $\rightarrow$ $\frac{9}{59}$ sick
59 people

Chi-square distribution + Test

numbers (height, time) $\rightarrow$ continous value

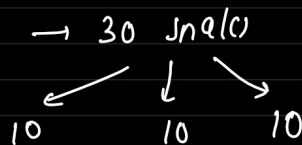
categorical $\rightarrow$?

$\rightarrow$ color, flavour, days of the week, yes/no

To handle categories, we use chi-square Test (χ^2)

(1) Expected v observation

fruit snack $\rightarrow$ 3 flavour (equal amount)



expectation:

observation: —

18 Mango
7 pineapple $\rightarrow$ } $\rightarrow$ expected $\rightarrow$ chi-square
5 kiwi

to find difference between expected & observed

df = total no of categories - 1

$$= 3 - 1$$

$$= 2$$

Formula: $\chi^2 = \sum \frac{(O - E)^2}{E}$

category	observed	expected	$(O-E)$	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Mango	18	10	8	64	$\frac{64}{10} = 6.4$
pineapple	7	10	-3	9	$\frac{9}{10} = 0.9$
Kiwi	5	10	-5	25	$\frac{25}{10} = 2.5$

step 1: Sum of $\sum \frac{(O-E)^2}{E}$

$$= 6.4 + 0.9 + 2.5$$

$$= 9.8$$

step 2: Find DOF

$$df = n - 1$$

$$= 3 - 1$$

$$= 2$$

The verdict :

$$df = 2, \alpha = 0.05, \text{critical value} =$$

$$\text{our result} = 9.8$$

bigger the gap
↓
too random to be
luck



critical value: This is the point where the gap becomes "too big to be luck"

small χ^2 : Means observed is very close to expected
: H_0 is likely true

large χ^2 : Means observed is a very far from expected
: H_0 is likely a lie

$\chi^2_{\text{stat}} \rightarrow$ observed

χ^2_{crit} $\rightarrow$ expected

observed $>$ expected

9.8 $>$ 5.991

Yes, $9.8 > 5.999$, we reject H_0 (Null Hypothesis)

X

starts - completed